

Incremental Subscene Updates for Unconstrained Online Dynamic Scene Reconstruction via 3D Gaussian Splatting in Fixed Multi-Camera Setups

Thesis Proposal (now obsolete)

submitted by

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Abstract

Online dynamic scene reconstruction via 3D Gaussian Splatting has seen comparatively little research, and existing works typically impose strict constraints on scene geometry. To address this limitation, the proposed thesis introduces a novel algorithm that incrementally updates the full 3D scene exclusively through small subscenes learned from frame-to-frame pixel changes. The focus lies on algorithmic feasibility and on performance evaluation in real-time contexts.

1 Introduction

3D Gaussian Splatting (3DGS), conceived in 2023, is a young AI technology in the field of 3D reconstruction which is itself part of the wider field of Computer Graphics. Technologies in the field of 3D reconstruction create a 3D scene from a set of 2D input imagery. They consist of a reconstruction part and a part rendering new views from the reconstructed scene (novel-view synthesis).

Before 3DGS, learning a 3D scene was done with large neural networks which learned an implicit radiance field representation¹ (Neural Radiance Fields, NeRFs). Training one scene usually took several days and novel-view synthesis reached very low numbers of frames per second (FPS) at best. It was not possible to be used for real-time rendering.

3DGS does not use neural networks as it learns an explicit radiance field representation. As its base primitive it makes use of 3D Gaussians, differentiable, ellipsoid fuzzy blobs, which adjust their position, color, shape, and opacity during the training procedure. The training procedure works with standard backpropagation via gradient descent. Using these base primitives as well as state-of-the-art utilization of modern GPUs enabled

¹An implicit radiance field is a continuous function that maps coordinates/points and a viewing angle to a color and opacity. An explicit radiance field limits the amount of points.

learning 3D scenes much quicker, in the order of minutes, while maintaining or improving NeRF image quality, as well as real-time novel-view synthesis (≥ 30 FPS) at 1080p resolution. The initial publication led to a massive surge of interest in the field.

Besides reconstructing static scenes as shown, dynamic scene reconstruction describes reconstructing moving/changing scenes from video feeds. A naive approach to having dynamic scenes in 3DGS would be to train a static scene for every frame. Even with training times taking only minutes, this would be a lot too slow for most purposes. That is why so-called 4DGS makes use of the fact that the previous scene has geometry very close in space to what the next frame requires and takes the previous geometry as the starting point for a much shorter training phase for the next iteration of the scene. While real-time novel-view synthesis remains possible in this scenario, the video imagery has to be captured first, then the dynamic scene has to be learned and stored using various encodings and then finally the scene can be replayed and new views rendered from it while the scene is playing. As the scene has to be captured beforehand, this is also called offline dynamic scene reconstruction.

There are only few attempts to achieve online dynamic scene reconstruction as it is framed as being computationally very demanding. Existing work therefore typically constrains the types of geometry allowed in a scene or adds other supporting and constraining methods like neural networks predicting the next frame or anatomic constructs guiding the 3D Gaussians. Unconstrained online dynamic scene reconstruction at this point in time takes several seconds or minutes per frame.

This is where the proposed thesis steps in. By using a fixed multi-camera setup rather than a moving one, we can extract much more information from each frame, especially when comparing to the previous frame (the frame delta). This way, we can identify subscenes of the full scene that are dynamic and only train those akin to offline dynamic scene reconstruction. As the amount of 3D Gaussians in the subscene will be significantly less than the full scene and the delta of two consecutive frames is much less than a full frame, this will compute much faster. The proposed thesis does not claim to achieve real-time dynamic scene reconstruction, but rather aims to validate and evaluate the proposed online algorithm which can then be developed further in future work.

Showing that there is a viable path for improved unconstrained online dynamic scene reconstruction is important for the field of 3D reconstruction and 3DGS as a technology as it would enable more subsequent research and transform or create downstream applications.

2 Background

After the informal introduction we will now formally introduce the various relevant technologies for the proposed thesis.

2.1 Brief Overview of 3D Gaussian Splatting (3DGS)

3DGS is a young AI technology in the field of 3D Reconstruction which is itself part of the wider Computer Graphics field. It was first published by Kerbl et al. at INRIA in France in 2023 [6] and has since experienced wide adoption and received critical acclaim across domains, having been cited more than 10,000 times [12, 11]. 3DGS solved issues previous technologies in 3D Reconstruction had, such as Neural Radiance Fields (NeRFs) [9], as for example being able to render 1080p resolution scenes in real-time (*Novel-View Synthesis*), making full use of modern GPGPUs, offering editability of the scene while still featuring very reasonable training times. 3DGS uses 3D Gaussians as the base primitive, a previously known data structure [20, 21], which can be described as ellipsoid fuzzy blobs that can take any ellipsoid shape from perfect spheres to thin and long shapes or flat and round shapes. They use a multi-dimensional normal distribution for their opacity distribution, being most opaque at the center and rapidly losing opacity towards the edges. Kerbl et al., among other achievements, made 3D Gaussians differentiable so they can be used in standard backpropagation via gradient descent. This makes 3DGS an explicit, learnable radiance field representation without neural networks.

2.2 3D Gaussian Splatting Algorithm

Regular 3DGS takes 2D input images and learns the pictured 3D scene from them. The training phase can run for 2-20 minutes, depending on scene size but already good results are available within seconds after the start of training [6]. A large 3D scene can have millions of 3D Gaussians and hundreds of input images. The input images are first converted into a point cloud and camera matrices (camera orientation in space etc.) using an algorithm called *Structure from Motion* [15] and the points of the point cloud become the origins for the initial 3D Gaussians before training. It is important to know that even initialising 3D Gaussians randomly still converges, but takes longer to train. Each forward and backward pass processes one random input image to adjust the 3D scene iteration by iteration until the loss between original input images and rendered images is small enough. The backward pass adjusts all attributes of a 3D Gaussian (position, opacity, shape, color) purely by the difference in pixel color between rendered and original image after the previous forward pass [6, 5].

I attended the AIG seminar „Artificial Intelligence - Visual Computing“ with 3DGS as the topic, covering the base concept and algorithm in much more detail.

2.3 Dynamic Scene Reconstruction

Dynamic scene reconstruction (as in „reconstructing a dynamic scene“) in general represents reconstruction of 3D scenes that change over time, which means the inputs are video feeds with a certain frame rate.

2.3.1 Camera Setups for Dynamic Scene Reconstruction

In general, dynamic scene reconstruction scenarios can be divided into single and multi-camera setups using fixed or mobile cameras. Mobile cameras add much to the general computational load as their position has to be estimated with every frame, which is why using fixed camera setups can help bound the load. This automatically limits the extent of the scene but comes naturally for e.g. confined indoor spaces.

Single camera setups can be found in 3DGS solutions to SLAM (Simultaneous Localization and Mapping) in robotics which use mobile cameras and live 3D telepresence with fixed cameras for example (e.g. [4, 10]). It is clear that with single cameras much of the environment has to be inferred which poses a very different problem.

Another option within this differentiation is so-called *event-based 3DGS* (e.g. [17]) which uses „event cameras“ (dynamic vision sensors) that only record changes in light intensity, without color. State-of-the-art solutions for SLAM or robotics in general use these kind of cameras allowing real-time 3D reconstruction of unconstrained scenes, however, at lower resolutions and without color for internal representations only.

Another option facilitating dynamic reconstruction are RGB-D cameras, which record depth information alongside color values (e.g. [4]). This however works only for small scenes or for short distances as larger scenes require Lidar-based sensors, which have their own restrictions and side-effects.

The proposed thesis makes use of fixed multi-camera setups with regular RGB cameras in order to be able to compare consecutive frames in a meaningful way. As the goal is to achieve online dynamic scene reconstruction, algorithmic details used in event-based SLAM could, however, also prove to be valuable input.

2.3.2 Offline Dynamic Scene Reconstruction

In the context of 3DGS as a technology, dynamic scene reconstruction is also called „4DGS“ or „Free Viewpoint Video“ (e.g. [19, 16, 18]). Training the initial scene is identical to regular 3DGS, but the following frames can rely on scene geometry from the previous frame, making training a lot faster as 3D Gaussians are already in the vicinity, with similar shape and color of the next frame’s content. The input imagery is still full 2D images and the full 3D scene is being considered for training.

The difference of the 3D scene between frames can be encoded into/learned by so-called *deformation fields*, small neural networks which allow transforming 3D Gaussians over time according to the field [16]. The other option is to encode the time component directly into the 3D Gaussian data structure, effectively transforming them into 4D Gaussians [18]. This means that 4DGS scenes are learned from pre-recorded imagery, then encoded and then later replayed and explored with novel-view synthesis. There is no real-time component during reconstruction², which is why the attribute „offline“ can be assigned to this scenario.

Recent works primarily focus on optimizing compression as a 4DGS scene can become very large quickly [7] or clustering 3D Gaussians by their motion [1]. The principle

²Novel-view synthesis remains real-time, as with regular 3DGS

algorithm of using the previous state of the scene as the starting point for the next frame has remained unchanged since inception.

2.3.3 Online and Real-Time Dynamic Scene Reconstruction

Contrary to 4DGS scenes, online dynamic scene reconstruction (also called „3DGS Live Streaming“, „Live 3DGS“, or „3DGS Live Video“³) describes concurrent reconstruction and novel-view synthesis of incoming video frames, usually after initial regular training of a base 3DGS scene. Online dynamic scene reconstruction does not have to be real-time, but it can be, which means that real-time dynamic scene reconstruction represents the strictest class in dynamic scene reconstruction.

In both cases, a set of new input imagery is converted into the next iteration of the full 3D scene using various algorithms on the fly. There is no need for deformation fields or 4D Gaussians, even though they have been used in this context to store the dynamic scene for later.

A problem for both, offline and online dynamic scene reconstruction, is occlusion handling. There will inevitably be parts in the scene that have not been captured by any camera as other geometry completely obstructs the views. For these regions, there are essentially three possibilities:

- Leave as is. The most honest approach.
- Use AI priors to fill the region with geometry that makes the most sense.
- Use statistical solutions to fill the region.

Constrained Solutions Single-camera setups described in Chapter 2.3.1 naturally have to use some kind of prediction mechanism to show what is behind the primary geometry the camera can capture. A prediction is always a constraint as only those things can be predicted which have been trained beforehand. The same applies to skeletal models (usually of humans) or immutable static backgrounds as for example in [3]. Using such constraints and large processing pipelines with corresponding GPU hardware makes real-time dynamic scene reconstruction possible - but such scenes are severely limited in content, amongst others.

Unconstrained Solutions Unconstrained online dynamic scene reconstruction has been possible as early as 2023 [8] when 3D Gaussians were modelled as physical particles. Later frameworks like 3DGStream [14] and HiCom [2] advanced this to achieve a much lower error rate as with [8] but training one frame would still take seconds to minutes. This also means that real-time dynamic scene reconstruction has so far not been achieved for unconstrained/arbitrary scenes. This thesis proposal tries to take one step towards reaching this goal.

³In 3DGS research, it seems common for authors to reinvent terminology.

3 Problem Description

The proposed thesis is about formulating and evaluating a novel approach to online dynamic scene reconstruction via 3DGS that only looks at the subset of 3D Gaussians - the *subscene* - that is being altered in comparison to the previous frame. Such a subscene contains only a fraction of all Gaussians and would therefore be much faster to compute.

In a first step, though, the space covered by the fixed multi-camera setup is trained as a regular 3DGS scene which functions as the base for subsequent updates, as with 4DGS.

The input imagery for a subscene are frame-to-frame pixel changes from the new frame to the previous frame (for each camera). This is explicitly not a difference image, as the input image would contain pixels that did change over two consecutive frames holding their changed RGB value. The rest of the pixels would be encoded as 'unset' and be ignored during subscene reconstruction.

During the forward pass of subscene reconstruction, only the set pixels (the ones not to be ignored) are considered and the loss function only compares the set pixels in rendered and input images before heading into the backward pass. Also in the backward pass, only Gaussians behind set pixels are considered - this is provided by 3DGS out of the box. All other Gaussians would not be touched during gradient descent („frozen“).

The algorithm is expected to converge very quickly as

- the existing geometry is spatially close to what the new imagery of a new frame requires.
- solutions for new geometry entering the scene require spawning new 3D Gaussians at the correct position. This can be achieved by finding the intersection of changed pixels from at least two input images and mapping those to points in 3D space where the Gaussians can be spawned. We would spawn a larger amount of empty Gaussians in this spot and prune the ones that remained empty after each new frame. So there would be some kind of secondary pruning operation after the regular pruning during training of the base scene.
- geometry removed/leaving the scene will be handled in the regular way as the pixel color will then show the color behind the geometry that was removed. This geometry will have been known before and the algorithm should only remove the 3D Gaussians that were in front of the one that provided the color behind.

This means that the training algorithm for the subscene is to a large extent identical to regular 3DGS/4DGS, but using a mask on pixels and Gaussians to be considered.

This process could be sped up if Gaussians contained motion vectors making predicting the Gaussians' next positions more precise, reducing the amount of training iterations.

This subscene approach in a fixed multi-camera setup ~~has never been investigated but~~ looks very promising to at least reach a low number of FPS which can then later – in subsequent work – be refined to run at real-time speeds („First make it work, then make it work fast“). Refer to Chapter 5 for an explanation of the strikethrough segment above.

Other Challenges

- Implementing such a niche idea cannot be done from scratch. As there are several open source repositories available, some time will have to be spent finding a suitable repository to base the implementation of the proposed thesis off of.
- In order to validate the implementation there are at least two datasets publicly available that have been used in previous papers for validation purposes. Alternatives would mean creating a scene from scratch through software like Blender.
- If there's time left, how could motion vectors be incorporated into the Gaussians?
- Maybe adjusting existing Gaussians and creating new Gaussians has to/should be done in two passes? How to classify existing vs. new geometry?

4 Goals of the thesis

The primary goal is to develop and evaluate the mentioned algorithm which identifies and trains subscenes. The goal is not to have it run in real-time, which is out of scope for a master thesis. I will implement it in PyTorch based off of existing repositories. I suggest the following rough thesis structure:

Literature and Introduction Extensive introduction into 3D Gaussian Splatting. I will use my seminar paper as a base for the introduction. This will be followed by an extensive introduction to dynamic scene reconstruction. As there aren't many papers dealing with multi-camera setups or even online dynamic scene reconstruction (around 10-12 relevant ones), I can introduce each paper, also categorizing them. This section can contain up to 50 pages.

Problem description/developing the algorithm Describing the problem the thesis deals with as well as developing the algorithm in detail on a theoretical basis. This section will be around 20 pages.

Overview of Available Code repositories A rather short overview of existing 3DGS/4DGS code repositories, weighing their benefits. In the end, I'll pick one to base the implementation off of. Around 5 pages.

Simulated Environment for a Fixed Multi-Camera Setup Some existing papers provide reference scenes recorded by multi-camera setups. I will examine which ones exist, evaluate their suitability and pinpoint possible pitfalls etc. These existing recordings can then be replayed and used as if they were live feeds. Around 5 pages.

Implementation The actual code will go into the appendix, but here I will compare the theoretical foundations of the algorithm from the 'Problem description/developing the algorithm' chapter with the implementation details. Around 10 pages.

Evaluation Here, the implementation is evaluated against a real-time context. As I already mentioned, the implementation is most likely not going to be real-time, so I will evaluate how close it is coming to real-time running, where obstacles and improvements lie, if there is certain geometry that is harder to use in a subscene, and if image quality can be maintained etc. The concluding parts, of course, are a discussion about future work and a summary. Around 20 pages.

5 Why This Proposal Has Become Obsolete

Two days before turning this proposal in, I ran another search for papers dealing with my proposed idea. As I had so far not found one paper dealing with the approach of using frame deltas, I had been skeptical if my searches had been thorough enough. And on September 22nd I found a preprint called *DecoGS* [13] which had only been uploaded seven days before. After reading, it became very clear that the authors had the very same ideas, pretty much down to the very details.

They successfully implemented the proposed ideas and their evaluation shows that image quality is superior to previous solutions like 3DGStream. Training times could however not be reduced. As an accompanying repository has not yet been made available to the public, we can only guess as to why this is.

6 About me

I am 43, from Berlin and I studied Diplom-Informatik at Karlsruhe Institute of Technology, graduating in 2010. After that I spent around twelve years working as a software engineer, mostly on the data and backend sides of projects. I decided to return to enrolling at a university and studying Data Science mostly because of two long-term diseases that prevented me from performing my previous jobs (including a PhD program) but also from the felt need to get up to speed on more recent topics in Computer Science. As an experienced software developer with a Diplom in CS, I focussed heavily on all modules offered by AIG as they offered the most added value compared to „regular“ CS modules. Keen on computer visuals, I was happy to learn about 3D Gaussian Splatting (3DGS) in late 2025 and have since tried to focus on this area as a thesis topic. 3DGS offers the challenge I seek at the intersection of advanced Computer Science and ML/AI. The idea of the thesis topic stems from an older thought experiment of mine, which saw true 3D video broadcasts of live events become possible.

I attended the seminar „Artificial Intelligence - Visual Computing“ in the 2026 summer term, actually proposing 3DGS as a seminar topic early on. The seminar paper I then wrote on 3DGS forms an important knowledge base of this proposal and the proposed master thesis.

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